

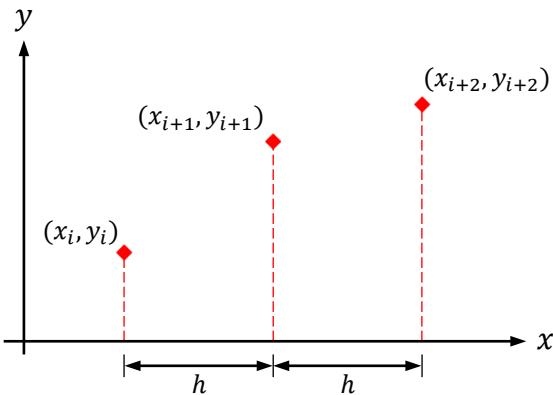
Newton-Cotes Formulas of Closed Type

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Simpson's $1/3$ Rule

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Simpson's 1/3 Rule

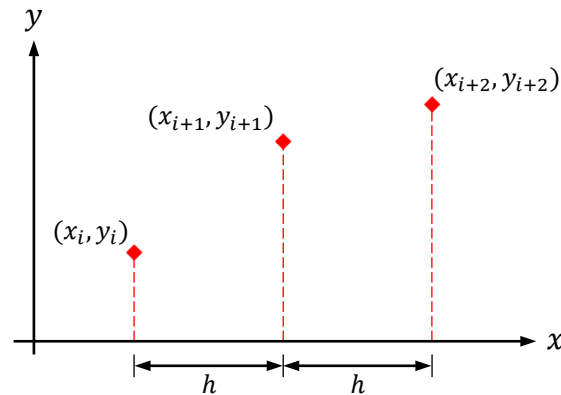


Newton-Cotes Formulas of Closed Type

Simpson's 1/3 Rule

$$n = 2$$

$$f(x) \approx p_2(x)$$

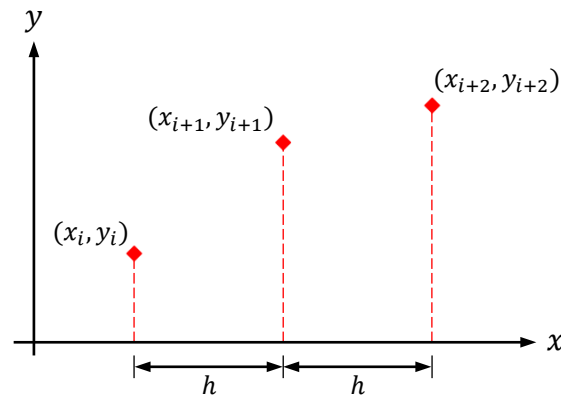


Newton-Cotes Formulas of Closed Type

Simpson's 1/3 Rule

 $n = 2$

$$f(x) \approx p_2(x) = L_0^{(2)}(x) \cdot f(x_i) + L_1^{(2)}(x) \cdot f(x_{i+1}) + L_2^{(2)}(x) \cdot f(x_{i+2})$$

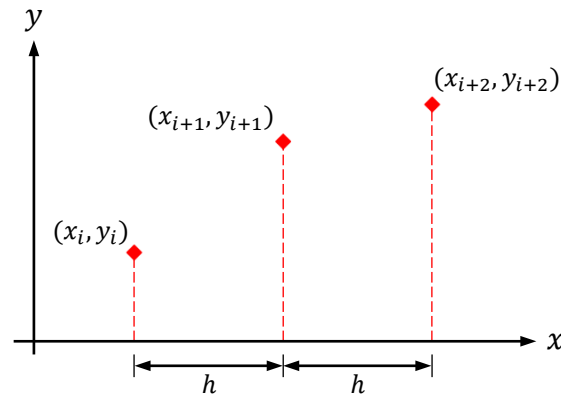


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Simpson's 1/3 Rule

$n = 2$

$$\begin{aligned}
 f(x) &\approx p_2(x) = L_0^{(2)}(x) \cdot f(x_i) + L_1^{(2)}(x) \cdot f(x_{i+1}) + L_2^{(2)}(x) \cdot f(x_{i+2}) \\
 &= \frac{x-x_{i+1}}{x_i-x_{i+1}} \cdot \frac{x-x_{i+2}}{x_i-x_{i+2}} \cdot f(x_i) + \frac{x-x_i}{x_{i+1}-x_i} \cdot \frac{x-x_{i+2}}{x_{i+1}-x_{i+2}} \cdot f(x_{i+1}) \\
 &\quad + \frac{x-x_i}{x_{i+2}-x_i} \cdot \frac{x-x_{i+1}}{x_{i+2}-x_{i+1}} \cdot f(x_{i+2})
 \end{aligned}$$

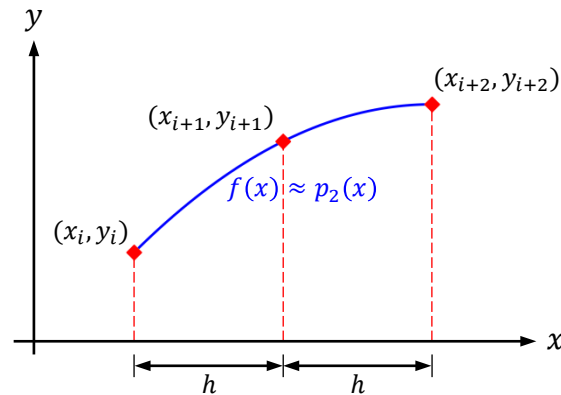


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 $n = 2$

$$\begin{aligned} f(x) &\approx p_2(x) = L_0^{(2)}(x) \cdot f(x_i) + L_1^{(2)}(x) \cdot f(x_{i+1}) + L_2^{(2)}(x) \cdot f(x_{i+2}) \\ &= \frac{x-x_{i+1}}{x_i-x_{i+1}} \cdot \frac{x-x_{i+2}}{x_i-x_{i+2}} \cdot f(x_i) + \frac{x-x_i}{x_{i+1}-x_i} \cdot \frac{x-x_{i+2}}{x_{i+1}-x_{i+2}} \cdot f(x_{i+1}) \\ &\quad + \frac{x-x_i}{x_{i+2}-x_i} \cdot \frac{x-x_{i+1}}{x_{i+2}-x_{i+1}} \cdot f(x_{i+2}) \end{aligned}$$



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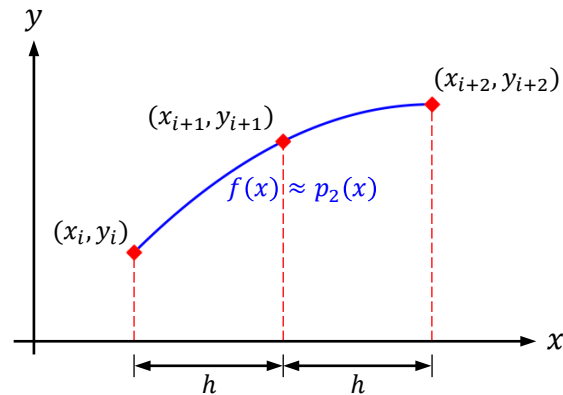
$n = 2$

$$f(x) \approx p_2(x) = L_0^{(2)}(x) \cdot f(x_i) + L_1^{(2)}(x) \cdot f(x_{i+1}) + L_2^{(2)}(x) \cdot f(x_{i+2})$$

$$= \frac{x-x_{i+1}}{x_i-x_{i+1}} \cdot \frac{x-x_{i+2}}{x_i-x_{i+2}} \cdot f(x_i) + \frac{x-x_i}{x_{i+1}-x_i} \cdot \frac{x-x_{i+2}}{x_{i+1}-x_{i+2}} \cdot f(x_{i+1})$$

$$+ \frac{x-x_i}{x_{i+2}-x_i} \cdot \frac{x-x_{i+1}}{x_{i+2}-x_{i+1}} \cdot f(x_{i+2})$$

$$\int_{x_i}^{x_{i+2}} f(x) \cdot dx \approx \int_{x_i}^{x_{i+2}} p_2(x) \cdot dx$$



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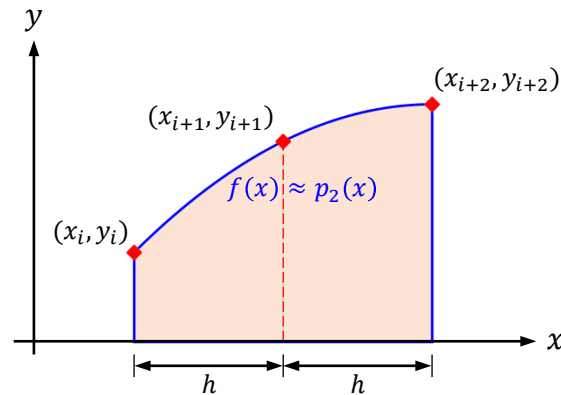
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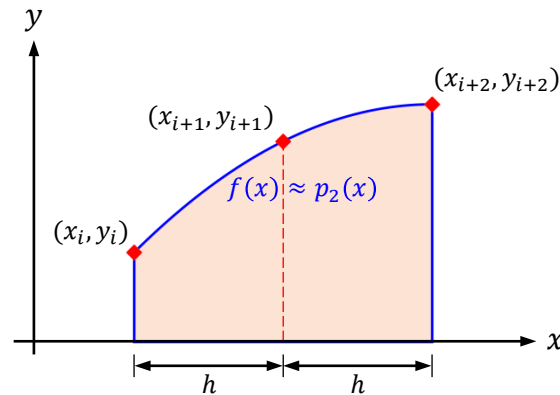
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$$\xi = \frac{x - x_i}{x_{i+1} - x_i} = \frac{x - x_i}{h}$$



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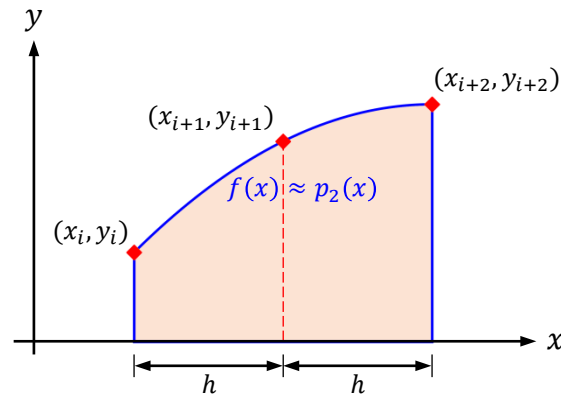
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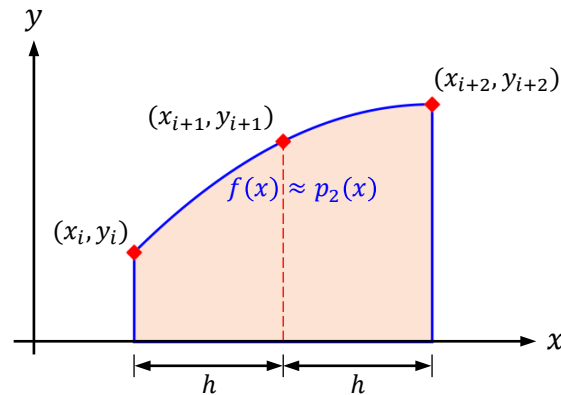
$$\int_{x_i}^{x_{i+2}} f(x) \cdot dx \approx \int_{x_i}^{x_{i+2}} p_2(x) \cdot dx$$

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$$x = x_i \rightarrow \xi = 0$$

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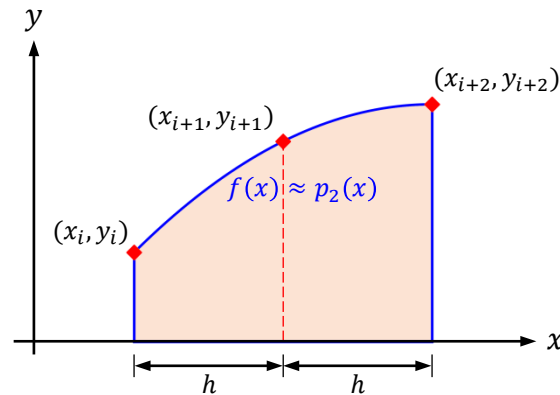
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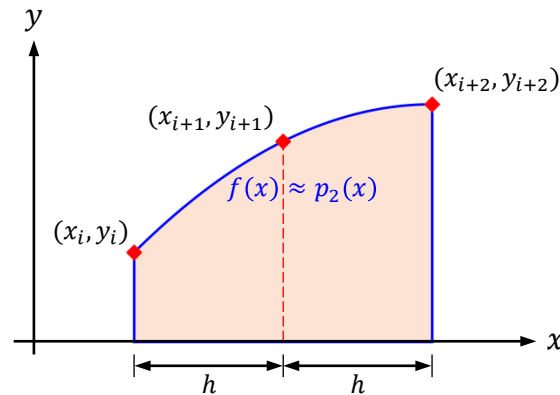
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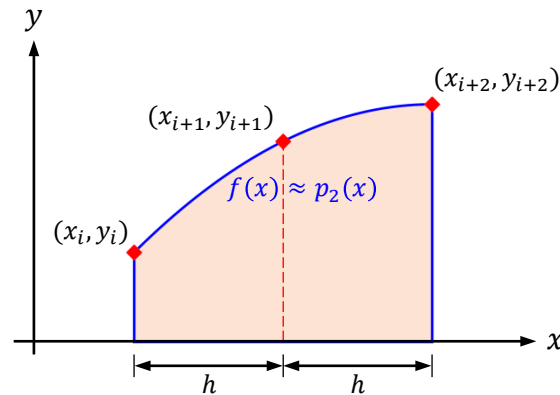
$$x = x_i \rightarrow \xi = 0$$

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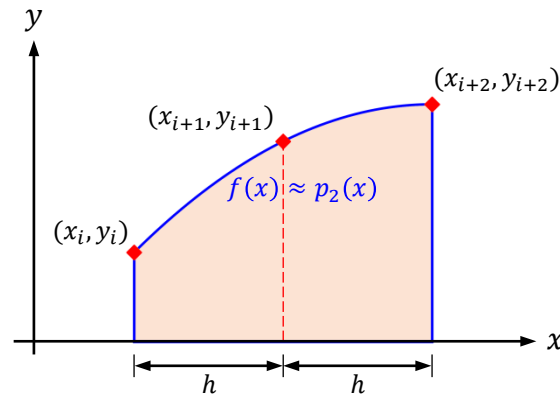
$$x = x_i \rightarrow \xi = 0$$

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$$\int_{x_i}^{x_{i+2}} \frac{x - x_i}{x_{i+1} - x_i} \cdot \frac{x - x_{i+2}}{x_{i+1} - x_{i+2}} \cdot dx = \int_0^2 \frac{h \cdot (\xi)}{h} \cdot \frac{h \cdot (\xi - 2)}{-h} \cdot (h \cdot d\xi) = \frac{4h}{3}$$



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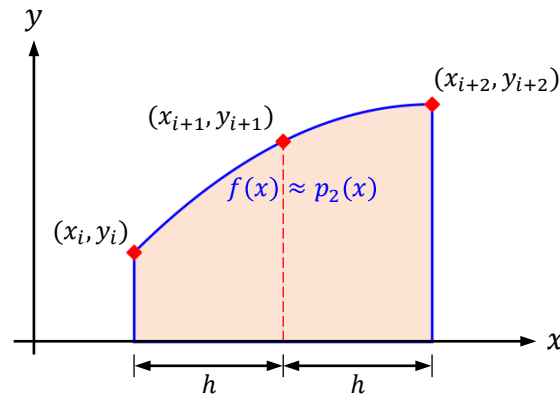
$$x = x_{i+1} \rightarrow \xi = 1$$

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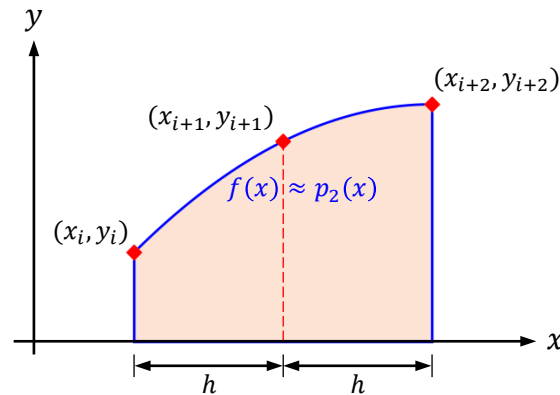
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$$+ \frac{x-x_i}{x_{i+2}-x_i} \cdot \frac{x-x_{i+1}}{x_{i+2}-x_{i+1}} \cdot f(x_{i+2})$$

$$\int_{x_i}^{x_{i+2}} f(x) \cdot dx \approx \int_{x_i}^{x_{i+2}} p_2(x) \cdot dx = \frac{h}{3} \cdot [f(x_i) + 4 \cdot f(x_{i+1}) + f(x_{i+2})]$$

$$\xi = \frac{x - x_i}{x_{i+1} - x_i} = \frac{x - x_i}{h} \rightarrow d\xi = \frac{1}{h} \cdot dx$$

$$x = x_i \rightarrow \xi = 0$$

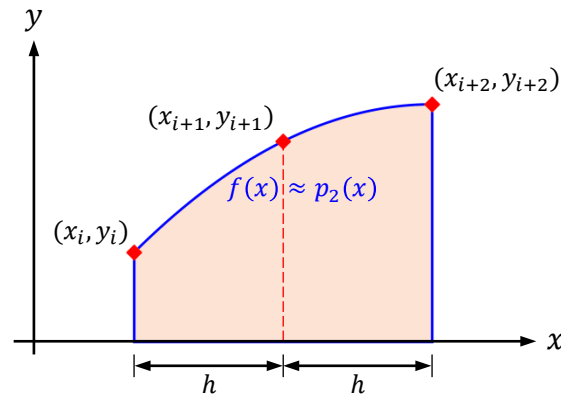
$$x = x_{i+1} \rightarrow \xi = 1$$

$$x = x_{i+2} \rightarrow \xi = 2$$

$$\int_{x_i}^{x_{i+2}} \frac{x - x_{i+1}}{x_i - x_{i+1}} \cdot \frac{x - x_{i+2}}{x_i - x_{i+2}} \cdot dx = \int_0^2 \frac{h \cdot (\xi - 1)}{-h} \cdot \frac{h \cdot (\xi - 2)}{-2h} \cdot (h \cdot d\xi) = \frac{h}{3}$$

$$\int_{x_i}^{x_{i+2}} \frac{x - x_i}{x_{i+1} - x_i} \cdot \frac{x - x_{i+2}}{x_{i+1} - x_{i+2}} \cdot dx = \int_0^2 \frac{h \cdot (\xi)}{h} \cdot \frac{h \cdot (\xi - 2)}{-h} \cdot (h \cdot d\xi) = \frac{4h}{3}$$

$$\int_{x_i}^{x_{i+2}} \frac{x - x_i}{x_{i+2} - x_i} \cdot \frac{x - x_{i+1}}{x_{i+2} - x_{i+1}} \cdot dx = \int_0^2 \frac{h \cdot (\xi)}{2h} \cdot \frac{h \cdot (\xi - 1)}{h} \cdot (h \cdot d\xi) = \frac{h}{3}$$



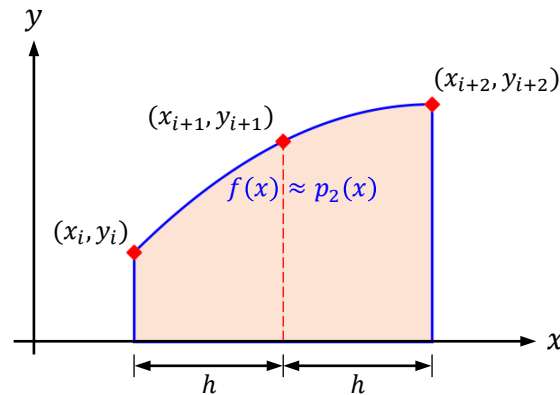
Newton-Cotes Formulas of Closed Type

Simpson's 1/3 Rule

$$n = 2$$

$$\begin{aligned} f(x) &\approx p_2(x) = L_0^{(2)}(x) \cdot f(x_i) + L_1^{(2)}(x) \cdot f(x_{i+1}) + L_2^{(2)}(x) \cdot f(x_{i+2}) \\ &= \frac{x-x_{i+1}}{x_i-x_{i+1}} \cdot \frac{x-x_{i+2}}{x_i-x_{i+2}} \cdot f(x_i) + \frac{x-x_i}{x_{i+1}-x_i} \cdot \frac{x-x_{i+2}}{x_{i+1}-x_{i+2}} \cdot f(x_{i+1}) \\ &\quad + \frac{x-x_i}{x_{i+2}-x_i} \cdot \frac{x-x_{i+1}}{x_{i+2}-x_{i+1}} \cdot f(x_{i+2}) \end{aligned}$$

$$\int_{x_i}^{x_{i+2}} f(x) \cdot dx \approx \int_{x_i}^{x_{i+2}} p_2(x) \cdot dx = \frac{h}{3} \cdot [f(x_i) + 4 \cdot f(x_{i+1}) + f(x_{i+2})]$$



$$\xi = \frac{x - x_i}{x_{i+1} - x_i} = \frac{x - x_i}{h} \rightarrow d\xi = \frac{1}{h} \cdot dx$$

$$x = x_i \rightarrow \xi = 0$$

$$x = x_{i+1} \rightarrow \xi = 1$$

$$x = x_{i+2} \rightarrow \xi = 2$$

$$\int_{x_i}^{x_{i+2}} \frac{x - x_{i+1}}{x_i - x_{i+1}} \cdot \frac{x - x_{i+2}}{x_i - x_{i+2}} \cdot dx = \int_0^2 \frac{h \cdot (\xi - 1)}{-h} \cdot \frac{h \cdot (\xi - 2)}{-2h} \cdot (h \cdot d\xi) = \frac{h}{3}$$

$$\int_{x_i}^{x_{i+2}} \frac{x - x_i}{x_{i+1} - x_i} \cdot \frac{x - x_{i+2}}{x_{i+1} - x_{i+2}} \cdot dx = \int_0^2 \frac{h \cdot (\xi)}{h} \cdot \frac{h \cdot (\xi - 2)}{-h} \cdot (h \cdot d\xi) = \frac{4h}{3}$$

$$\int_{x_i}^{x_{i+2}} \frac{x - x_i}{x_{i+2} - x_i} \cdot \frac{x - x_{i+1}}{x_{i+2} - x_{i+1}} \cdot dx = \int_0^2 \frac{h \cdot (\xi)}{2h} \cdot \frac{h \cdot (\xi - 1)}{h} \cdot (h \cdot d\xi) = \frac{h}{3}$$

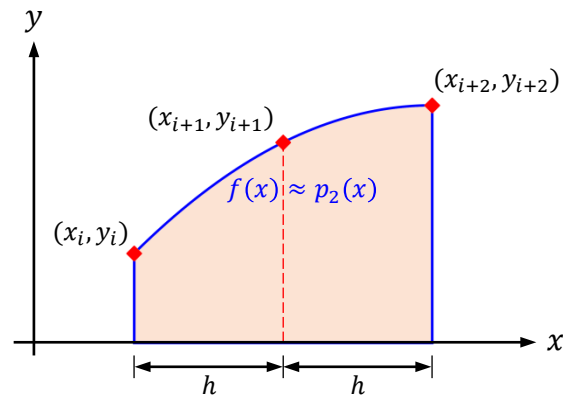
Newton-Cotes Formulas of Closed Type

Newton-Cotes Formulas of Closed Type

Simpson's $1/3$ Rule: The Local Truncation Error

Newton-Cotes Formulas of Closed Type

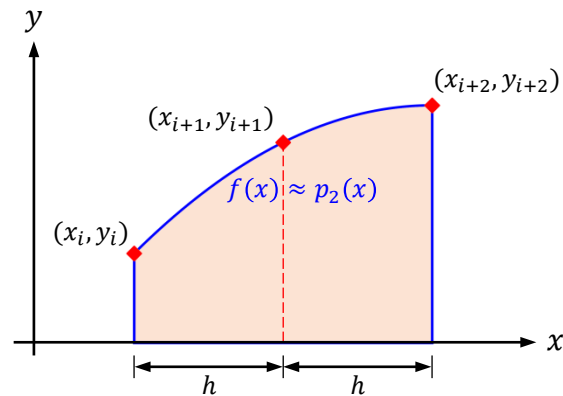
Simpson's 1/3 Rule: The Local Truncation Error



Newton-Cotes Formulas of Closed Type

Simpson's 1/3 Rule: The Local Truncation Error

$$f(x) = f(x_i) + (x - x_i) \cdot f'(x_i) + \frac{(x - x_i)^2}{2!} \cdot f''(x_i) + \frac{(x - x_i)^3}{3!} \cdot f^{(3)}(x_i) + \dots$$

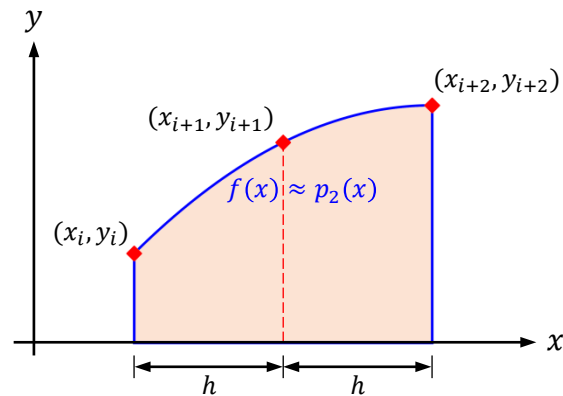


Newton-Cotes Formulas of Closed Type

Simpson's 1/3 Rule: The Local Truncation Error

$$f(x) = f(x_i) + (x - x_i) \cdot f'(x_i) + \frac{(x - x_i)^2}{2!} \cdot f''(x_i) + \frac{(x - x_i)^3}{3!} \cdot f^{(3)}(x_i) + \dots$$

$$\begin{aligned} \int_{x_i}^{x_{i+2}} f(x) \cdot dx &= (x_{i+2} - x_i) \cdot f(x_i) + \frac{(x_{i+2} - x_i)^2}{2!} \cdot f'(x_i) \\ &\quad + \frac{(x_{i+2} - x_i)^3}{3!} \cdot f''(x_i) + \frac{(x_{i+2} - x_i)^4}{4!} \cdot f^{(3)}(x_i) + \dots \end{aligned}$$

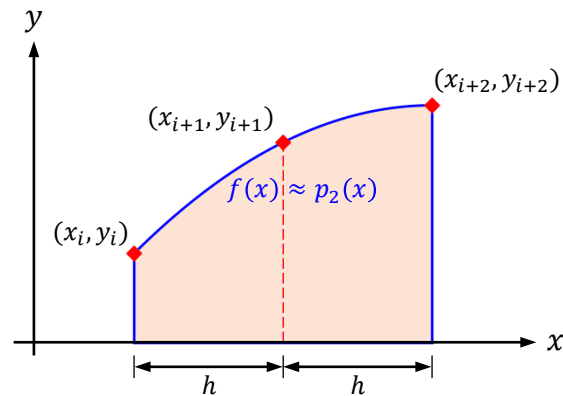


Newton-Cotes Formulas of Closed Type

Simpson's 1/3 Rule: The Local Truncation Error

$$f(x) = f(x_i) + (x - x_i) \cdot f'(x_i) + \frac{(x - x_i)^2}{2!} \cdot f''(x_i) + \frac{(x - x_i)^3}{3!} \cdot f^{(3)}(x_i) + \dots$$

$$\int_{x_i}^{x_{i+2}} f(x) \cdot dx = (2h) \cdot f(x_i) + \frac{(2h)^2}{2} \cdot f'(x_i) + \frac{(2h)^3}{6} \cdot f''(x_i) + \frac{(2h)^4}{24} \cdot f^{(3)}(x_i) + \frac{(2h)^5}{120} \cdot f^{(4)}(x_i) + \dots$$



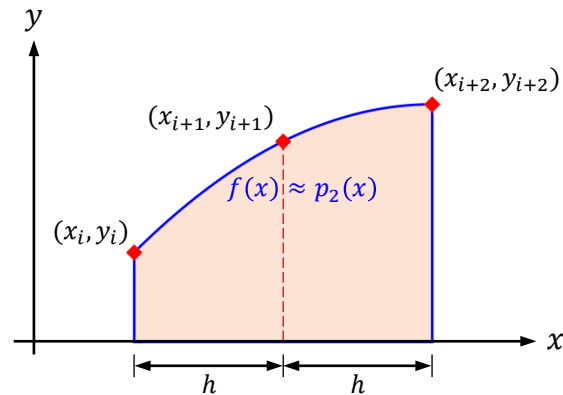
Newton-Cotes Formulas of Closed Type

Simpson's 1/3 Rule: The Local Truncation Error

$$f(x) = f(x_i) + (x - x_i) \cdot f'(x_i) + \frac{(x - x_i)^2}{2!} \cdot f''(x_i) + \frac{(x - x_i)^3}{3!} \cdot f^{(3)}(x_i) + \dots$$

$$\int_{x_i}^{x_{i+2}} f(x) \cdot dx = (2h) \cdot f(x_i) + \frac{(2h)^2}{2} \cdot f'(x_i) + \frac{(2h)^3}{6} \cdot f''(x_i) + \frac{(2h)^4}{24} \cdot f^{(3)}(x_i) + \frac{(2h)^5}{120} \cdot f^{(4)}(x_i) + \dots$$

$$R_{[x_i, x_{i+2}]}^{(2)} = \int_{x_i}^{x_{i+2}} f(x) \cdot dx - \int_{x_i}^{x_{i+2}} p_2(x) \cdot dx$$



Newton-Cotes Formulas of Closed Type

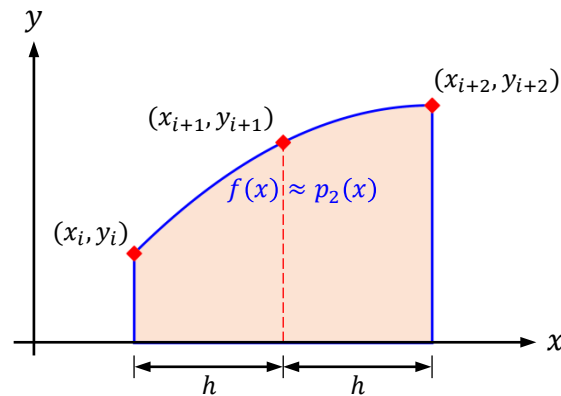
Simpson's 1/3 Rule: The Local Truncation Error

$$f(x) = f(x_i) + (x - x_i) \cdot f'(x_i) + \frac{(x - x_i)^2}{2!} \cdot f''(x_i) + \frac{(x - x_i)^3}{3!} \cdot f^{(3)}(x_i) + \dots$$

$$\int_{x_i}^{x_{i+2}} f(x) \cdot dx = (2h) \cdot f(x_i) + \frac{(2h)^2}{2} \cdot f'(x_i) + \frac{(2h)^3}{6} \cdot f''(x_i) + \frac{(2h)^4}{24} \cdot f^{(3)}(x_i) + \frac{(2h)^5}{120} \cdot f^{(4)}(x_i) + \dots$$

$$R_{[x_i, x_{i+2}]}^{(2)} = \int_{x_i}^{x_{i+2}} f(x) \cdot dx - \int_{x_i}^{x_{i+2}} p_2(x) \cdot dx = 2h \cdot f(x_i) + 2h^2 \cdot f'(x_i) + \frac{4h^3}{3} \cdot f''(x_i) + \frac{2h^4}{3} \cdot f^{(3)}(x_i) + \frac{4h^5}{15} \cdot f^{(4)}(x_i) + \dots$$

$$- \frac{h}{3} \cdot f(x_i) - \frac{4h}{3} \cdot f(x_{i+1}) - \frac{h}{3} \cdot f(x_{i+2})$$



Newton-Cotes Formulas of Closed Type

Simpson's 1/3 Rule: The Local Truncation Error

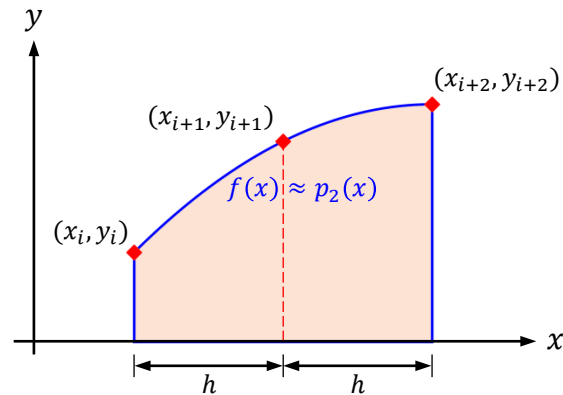
$$f(x) = f(x_i) + (x - x_i) \cdot f'(x_i) + \frac{(x - x_i)^2}{2!} \cdot f''(x_i) + \frac{(x - x_i)^3}{3!} \cdot f^{(3)}(x_i) + \dots$$

$$\int_{x_i}^{x_{i+2}} f(x) \cdot dx = (2h) \cdot f(x_i) + \frac{(2h)^2}{2} \cdot f'(x_i) + \frac{(2h)^3}{6} \cdot f''(x_i) + \frac{(2h)^4}{24} \cdot f^{(3)}(x_i) + \frac{(2h)^5}{120} \cdot f^{(4)}(x_i) + \dots$$

$$R_{[x_i, x_{i+2}]}^{(2)} = \int_{x_i}^{x_{i+2}} f(x) \cdot dx - \int_{x_i}^{x_{i+2}} p_2(x) \cdot dx = 2h \cdot f(x_i) + 2h^2 \cdot f'(x_i) + \frac{4h^3}{3} \cdot f''(x_i) + \frac{2h^4}{3} \cdot f^{(3)}(x_i) + \frac{4h^5}{15} \cdot f^{(4)}(x_i) + \dots$$

$$- \frac{h}{3} \cdot f(x_i) - \frac{4h}{3} \cdot \left[f(x_i) + h \cdot f'(x_i) + \frac{h^2}{2} \cdot f''(x_i) + \frac{h^3}{6} \cdot f^{(3)}(x_i) + \frac{h^4}{24} \cdot f^{(4)}(x_i) + \dots \right]$$

$$- \frac{h}{3} \cdot \left[f(x_i) + (2h) \cdot f'(x_i) + \frac{(2h)^2}{2} \cdot f''(x_i) + \frac{(2h)^3}{6} \cdot f^{(3)}(x_i) + \frac{(2h)^4}{24} \cdot f^{(4)}(x_i) + \dots \right]$$



Newton-Cotes Formulas of Closed Type

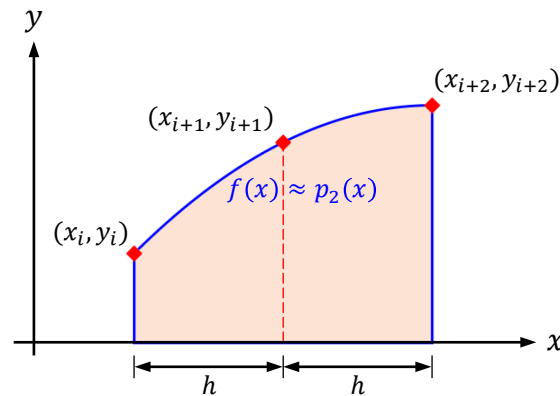
Simpson's 1/3 Rule: The Local Truncation Error

$$f(x) = f(x_i) + (x - x_i) \cdot f'(x_i) + \frac{(x - x_i)^2}{2!} \cdot f''(x_i) + \frac{(x - x_i)^3}{3!} \cdot f^{(3)}(x_i) + \dots$$

$$\int_{x_i}^{x_{i+2}} f(x) \cdot dx = (2h) \cdot f(x_i) + \frac{(2h)^2}{2} \cdot f'(x_i) + \frac{(2h)^3}{6} \cdot f''(x_i) + \frac{(2h)^4}{24} \cdot f^{(3)}(x_i) + \frac{(2h)^5}{120} \cdot f^{(4)}(x_i) + \dots$$

$$R_{[x_i, x_{i+2}]}^{(2)} = \int_{x_i}^{x_{i+2}} f(x) \cdot dx - \int_{x_i}^{x_{i+2}} p_2(x) \cdot dx = 2h \cdot f(x_i) + 2h^2 \cdot f'(x_i) + \frac{4h^3}{3} \cdot f''(x_i) + \frac{2h^4}{3} \cdot f^{(3)}(x_i) + \frac{4h^5}{15} \cdot f^{(4)}(x_i) + \dots$$

$$- 2h \cdot f(x_i) - 2h^2 \cdot f'(x_i) - \frac{4h^3}{3} \cdot f''(x_i) - \frac{2h^4}{3} \cdot f^{(3)}(x_i) - \frac{5h^5}{18} \cdot f^{(4)}(x_i) - \dots$$



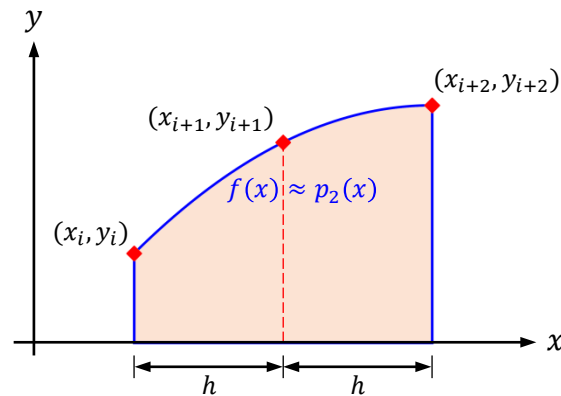
Newton-Cotes Formulas of Closed Type

Simpson's 1/3 Rule: The Local Truncation Error

$$f(x) = f(x_i) + (x - x_i) \cdot f'(x_i) + \frac{(x - x_i)^2}{2!} \cdot f''(x_i) + \frac{(x - x_i)^3}{3!} \cdot f^{(3)}(x_i) + \dots$$

$$\int_{x_i}^{x_{i+2}} f(x) \cdot dx = (2h) \cdot f(x_i) + \frac{(2h)^2}{2} \cdot f'(x_i) + \frac{(2h)^3}{6} \cdot f''(x_i) + \frac{(2h)^4}{24} \cdot f^{(3)}(x_i) + \frac{(2h)^5}{120} \cdot f^{(4)}(x_i) + \dots$$

$$R_{[x_i, x_{i+2}]}^{(2)} = \int_{x_i}^{x_{i+2}} f(x) \cdot dx - \int_{x_i}^{x_{i+2}} p_2(x) \cdot dx = -\frac{h^5}{90} \cdot f^{(4)}(x_i) + \dots$$



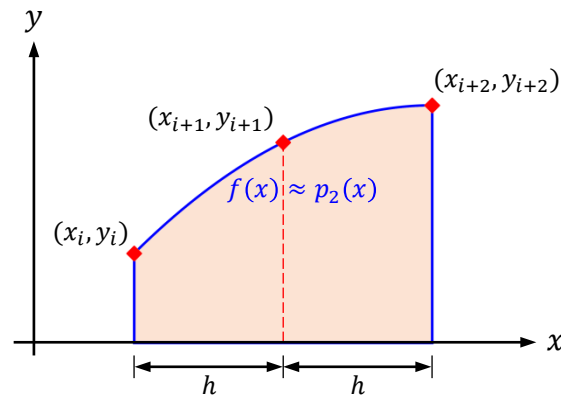
Newton-Cotes Formulas of Closed Type

Simpson's 1/3 Rule: The Local Truncation Error

$$f(x) = f(x_i) + (x - x_i) \cdot f'(x_i) + \frac{(x - x_i)^2}{2!} \cdot f''(x_i) + \frac{(x - x_i)^3}{3!} \cdot f^{(3)}(x_i) + \dots$$

$$\int_{x_i}^{x_{i+2}} f(x) \cdot dx = (2h) \cdot f(x_i) + \frac{(2h)^2}{2} \cdot f'(x_i) + \frac{(2h)^3}{6} \cdot f''(x_i) + \frac{(2h)^4}{24} \cdot f^{(3)}(x_i) + \frac{(2h)^5}{120} \cdot f^{(4)}(x_i) + \dots$$

$$\begin{aligned} R_{[x_i, x_{i+2}]}^{(2)} &= \int_{x_i}^{x_{i+2}} f(x) \cdot dx - \int_{x_i}^{x_{i+2}} p_2(x) \cdot dx = -\frac{h^5}{90} \cdot f^{(4)}(x_i) + \dots \\ &= -\frac{h^5}{90} \cdot f^{(4)}(\xi_{i+1}); \quad x_i \leq \xi_{i+1} \leq x_{i+2} \end{aligned}$$



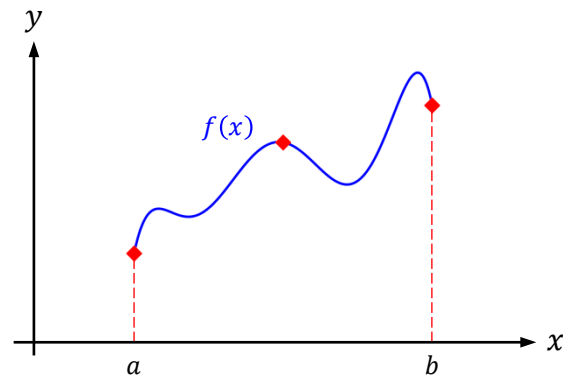
Newton-Cotes Formulas of Closed Type

Newton-Cotes Formulas of Closed Type

Simpson's $1/3$ Rule : The Global Truncation Error

Newton-Cotes Formulas of Closed Type

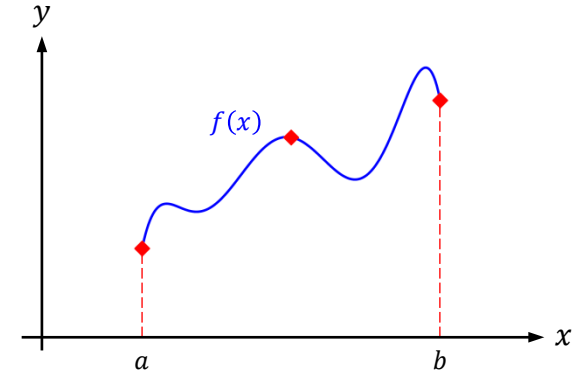
Simpson's 1/3 Rule : The Global Truncation Error



Newton-Cotes Formulas of Closed Type

Simpson's 1/3 Rule : The Global Truncation Error

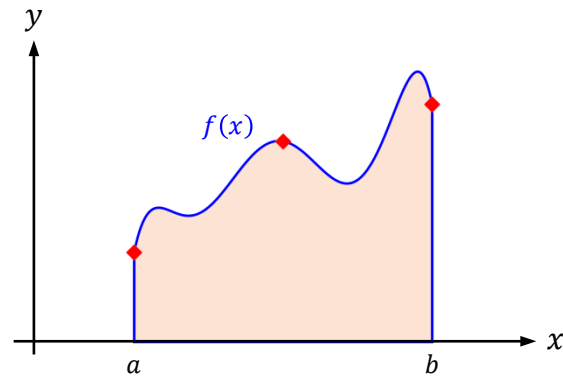
$$\int_a^b f(x) \cdot dx$$



Newton-Cotes Formulas of Closed Type

Simpson's 1/3 Rule : The Global Truncation Error

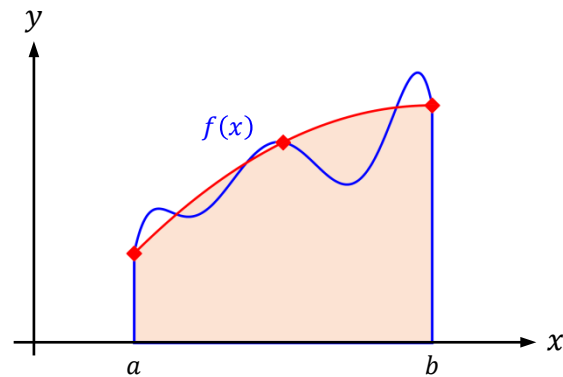
$$\int_a^b f(x) \cdot dx$$



Newton-Cotes Formulas of Closed Type

Simpson's 1/3 Rule : The Global Truncation Error

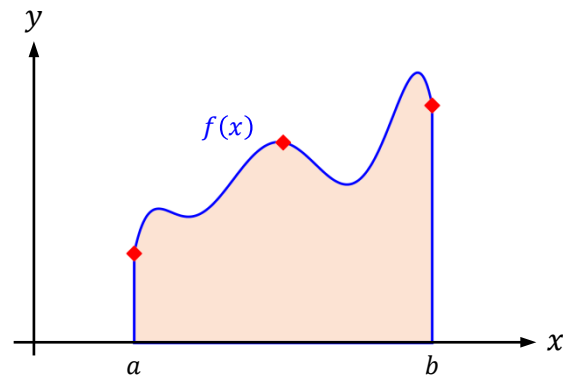
$$\int_a^b f(x) \cdot dx$$



Newton-Cotes Formulas of Closed Type

Simpson's 1/3 Rule : The Global Truncation Error

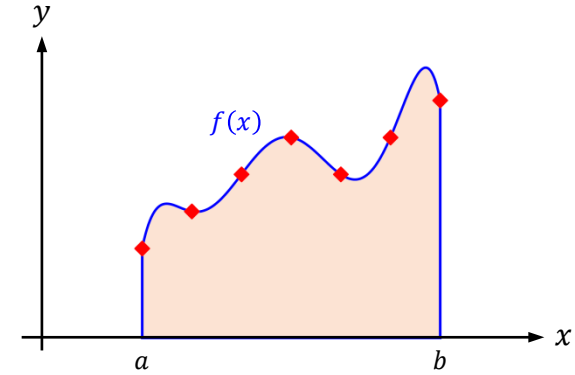
$$\int_a^b f(x) \cdot dx$$



Newton-Cotes Formulas of Closed Type

Simpson's 1/3 Rule : The Global Truncation Error

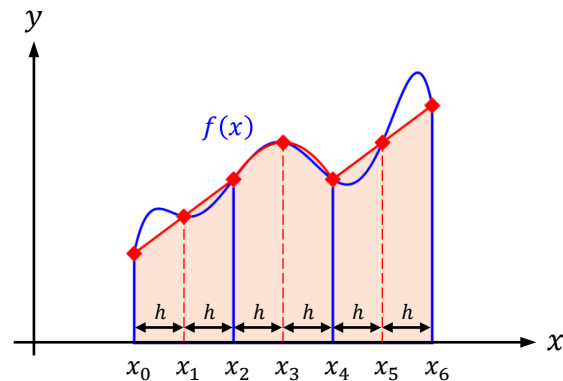
$$\int_a^b f(x) \cdot dx$$



Newton-Cotes Formulas of Closed Type

Simpson's 1/3 Rule : The Global Truncation Error

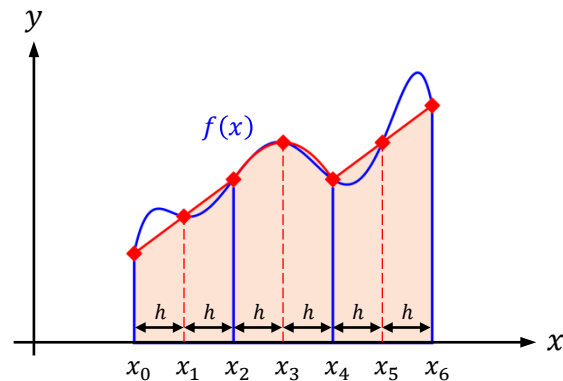
$$\int_a^b f(x) \cdot dx$$



Newton-Cotes Formulas of Closed Type

Simpson's 1/3 Rule : The Global Truncation Error

$$\begin{aligned} \int_a^b f(x) \cdot dx &= \int_a^{a+2h} f(x) \cdot dx + \int_{a+2h}^{a+4h} f(x) \cdot dx + \cdots + \int_{b-4h}^{b-2h} f(x) \cdot dx + \int_{b-2h}^b f(x) \cdot dx \\ &= \int_{x_0}^{x_2} f(x) \cdot dx + \int_{x_2}^{x_4} f(x) \cdot dx + \cdots + \int_{x_{N-4}}^{x_{N-2}} f(x) \cdot dx + \int_{x_{N-2}}^{x_N} f(x) \cdot dx \end{aligned}$$

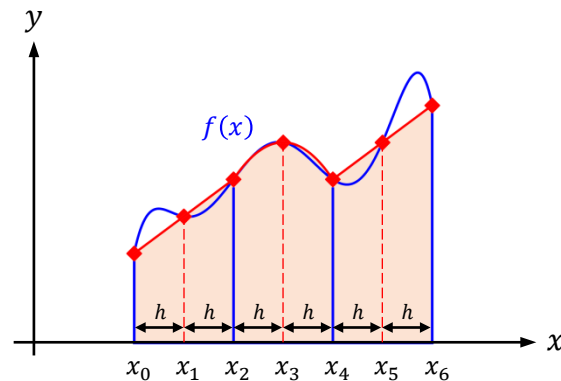


Newton-Cotes Formulas of Closed Type

Simpson's 1/3 Rule : The Global Truncation Error

$$\begin{aligned} \int_a^b f(x) \cdot dx &= \int_a^{a+2h} f(x) \cdot dx + \int_{a+2h}^{a+4h} f(x) \cdot dx + \cdots + \int_{b-4h}^{b-2h} f(x) \cdot dx + \int_{b-2h}^b f(x) \cdot dx \\ &= \int_{x_0}^{x_2} f(x) \cdot dx + \int_{x_2}^{x_4} f(x) \cdot dx + \cdots + \int_{x_{N-4}}^{x_{N-2}} f(x) \cdot dx + \int_{x_{N-2}}^{x_N} f(x) \cdot dx \end{aligned}$$

$$\begin{aligned} a &= x_0 \\ b &= x_N \end{aligned} \rightarrow h = \frac{b - a}{N}$$

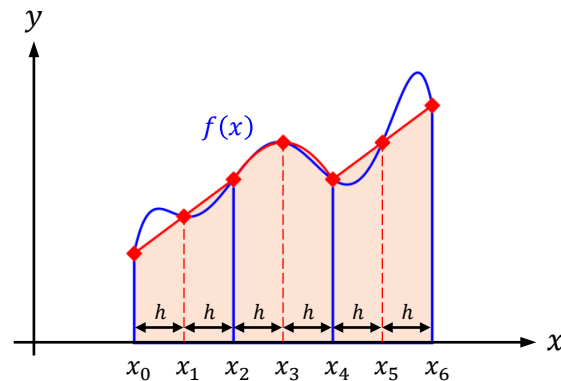


Newton-Cotes Formulas of Closed Type

Simpson's 1/3 Rule : The Global Truncation Error

$$\begin{aligned} \int_a^b f(x) \cdot dx &= \int_a^{a+2h} f(x) \cdot dx + \int_{a+2h}^{a+4h} f(x) \cdot dx + \cdots + \int_{b-4h}^{b-2h} f(x) \cdot dx + \int_{b-2h}^b f(x) \cdot dx \\ &= \int_{x_0}^{x_2} f(x) \cdot dx + \int_{x_2}^{x_4} f(x) \cdot dx + \cdots + \int_{x_{N-4}}^{x_{N-2}} f(x) \cdot dx + \int_{x_{N-2}}^{x_N} f(x) \cdot dx \end{aligned}$$

$$\begin{aligned} a &= x_0 \\ b &= x_N \end{aligned} \rightarrow h = \frac{b-a}{N} = x_1 - x_0 = x_2 - x_1 = \cdots = x_N - x_{N-1}$$



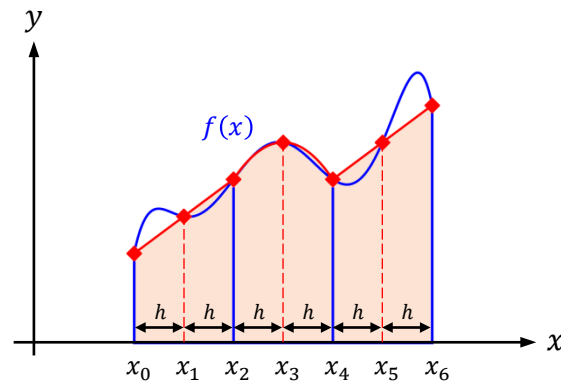
Newton-Cotes Formulas of Closed Type

Simpson's 1/3 Rule : The Global Truncation Error

$$\begin{aligned} \int_a^b f(x) \cdot dx &= \int_a^{a+2h} f(x) \cdot dx + \int_{a+2h}^{a+4h} f(x) \cdot dx + \cdots + \int_{b-4h}^{b-2h} f(x) \cdot dx + \int_{b-2h}^b f(x) \cdot dx \\ &= \int_{x_0}^{x_2} f(x) \cdot dx + \int_{x_2}^{x_4} f(x) \cdot dx + \cdots + \int_{x_{N-4}}^{x_{N-2}} f(x) \cdot dx + \int_{x_{N-2}}^{x_N} f(x) \cdot dx \end{aligned}$$

$$\begin{aligned} a &= x_0 \\ b &= x_N \end{aligned} \rightarrow h = \frac{b-a}{N} = x_1 - x_0 = x_2 - x_1 = \cdots = x_N - x_{N-1}$$

$$R_{[a,b]}^{(2)} = R_{[x_0,x_2]}^{(2)} + R_{[x_2,x_4]}^{(2)} + \cdots + R_{[x_{N-2},x_N]}^{(2)}$$



Newton-Cotes Formulas of Closed Type

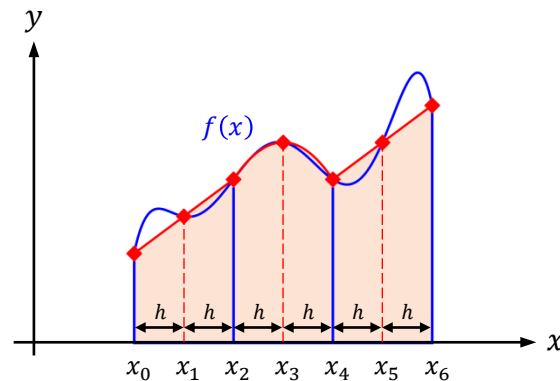
Simpson's 1/3 Rule : The Global Truncation Error

$$\begin{aligned} \int_a^b f(x) \cdot dx &= \int_a^{a+2h} f(x) \cdot dx + \int_{a+2h}^{a+4h} f(x) \cdot dx + \cdots + \int_{b-4h}^{b-2h} f(x) \cdot dx + \int_{b-2h}^b f(x) \cdot dx \\ &= \int_{x_0}^{x_2} f(x) \cdot dx + \int_{x_2}^{x_4} f(x) \cdot dx + \cdots + \int_{x_{N-4}}^{x_{N-2}} f(x) \cdot dx + \int_{x_{N-2}}^{x_N} f(x) \cdot dx \end{aligned}$$

$$\begin{aligned} a &= x_0 \\ b &= x_N \end{aligned} \rightarrow h = \frac{b-a}{N} = x_1 - x_0 = x_2 - x_1 = \cdots = x_N - x_{N-1}$$

$$R_{[a,b]}^{(2)} = R_{[x_0,x_2]}^{(2)} + R_{[x_2,x_4]}^{(2)} + \cdots + R_{[x_{N-2},x_N]}^{(2)} = -\frac{h^5}{90} \cdot f^{(4)}(\xi_1) - \frac{h^5}{90} \cdot f^{(4)}(\xi_2) - \cdots - \frac{h^5}{90} \cdot f^{(4)}(\xi_{N/2});$$

$$\begin{aligned} x_0 &\leq \xi_1 \leq x_2 \\ x_2 &\leq \xi_2 \leq x_4 \\ &\vdots \\ x_{N-2} &\leq \xi_{N/2} \leq x_N \end{aligned}$$



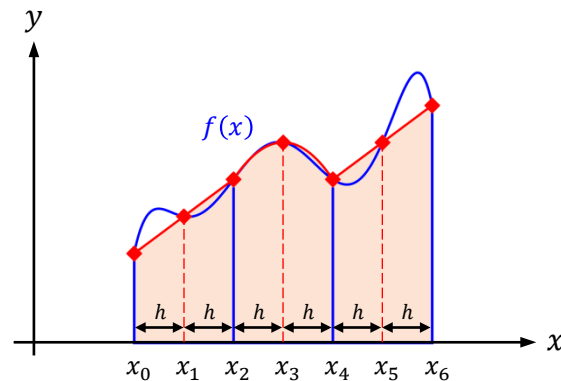
Newton-Cotes Formulas of Closed Type

Simpson's 1/3 Rule : The Global Truncation Error

$$\begin{aligned}
 \int_a^b f(x) \cdot dx &= \int_a^{a+2h} f(x) \cdot dx + \int_{a+2h}^{a+4h} f(x) \cdot dx + \cdots + \int_{b-4h}^{b-2h} f(x) \cdot dx + \int_{b-2h}^b f(x) \cdot dx \\
 &= \int_{x_0}^{x_2} f(x) \cdot dx + \int_{x_2}^{x_4} f(x) \cdot dx + \cdots + \int_{x_{N-4}}^{x_{N-2}} f(x) \cdot dx + \int_{x_{N-2}}^{x_N} f(x) \cdot dx
 \end{aligned}$$

$$\begin{aligned}
 a &= x_0 \\
 b &= x_N
 \end{aligned}
 \rightarrow h = \frac{b-a}{N} = x_1 - x_0 = x_2 - x_1 = \cdots = x_N - x_{N-1}$$

$$\begin{aligned}
 R_{[a,b]}^{(2)} &= R_{[x_0,x_2]}^{(2)} + R_{[x_2,x_4]}^{(2)} + \cdots + R_{[x_{N-2},x_N]}^{(2)} = -\frac{h^5}{90} \cdot f^{(4)}(\xi_1) - \frac{h^5}{90} \cdot f^{(4)}(\xi_2) - \cdots - \frac{h^5}{90} \cdot f^{(4)}(\xi_{N/2}); \\
 &= -\frac{\frac{N}{2} \cdot h^5}{90} \cdot f^{(4)}(\xi); \quad x_0 \leq \xi \leq x_N
 \end{aligned}$$



$$\begin{aligned}
 x_0 &\leq \xi_1 \leq x_2 \\
 x_2 &\leq \xi_2 \leq x_4 \\
 &\vdots \\
 x_{N-2} &\leq \xi_{N/2} \leq x_N
 \end{aligned}$$

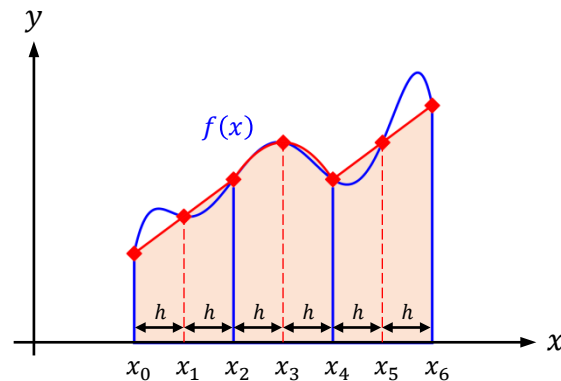
Newton-Cotes Formulas of Closed Type

Simpson's 1/3 Rule : The Global Truncation Error

$$\begin{aligned} \int_a^b f(x) \cdot dx &= \int_a^{a+2h} f(x) \cdot dx + \int_{a+2h}^{a+4h} f(x) \cdot dx + \cdots + \int_{b-4h}^{b-2h} f(x) \cdot dx + \int_{b-2h}^b f(x) \cdot dx \\ &= \int_{x_0}^{x_2} f(x) \cdot dx + \int_{x_2}^{x_4} f(x) \cdot dx + \cdots + \int_{x_{N-4}}^{x_{N-2}} f(x) \cdot dx + \int_{x_{N-2}}^{x_N} f(x) \cdot dx \end{aligned}$$

$$\begin{aligned} a &= x_0 \\ b &= x_N \end{aligned} \rightarrow h = \frac{b-a}{N} = x_1 - x_0 = x_2 - x_1 = \cdots = x_N - x_{N-1}$$

$$\begin{aligned} R_{[a,b]}^{(2)} &= R_{[x_0,x_2]}^{(2)} + R_{[x_2,x_4]}^{(2)} + \cdots + R_{[x_{N-2},x_N]}^{(2)} = -\frac{h^5}{90} \cdot f^{(4)}(\xi_1) - \frac{h^5}{90} \cdot f^{(4)}(\xi_2) - \cdots - \frac{h^5}{90} \cdot f^{(4)}(\xi_{N/2}); \\ &= -\frac{\frac{b-a}{N} \cdot h^5}{2 \cdot 90} \cdot f^{(4)}(\xi); \quad x_0 \leq \xi \leq x_N \end{aligned}$$



$$\begin{aligned} x_0 &\leq \xi_1 \leq x_2 \\ x_2 &\leq \xi_2 \leq x_4 \\ &\vdots \\ x_{N-2} &\leq \xi_{N/2} \leq x_N \end{aligned}$$

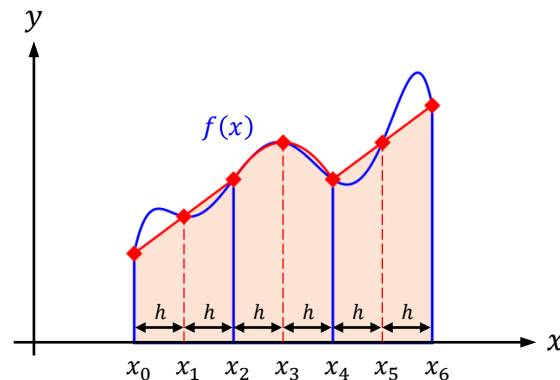
Newton-Cotes Formulas of Closed Type

Simpson's 1/3 Rule : The Global Truncation Error

$$\begin{aligned} \int_a^b f(x) \cdot dx &= \int_a^{a+2h} f(x) \cdot dx + \int_{a+2h}^{a+4h} f(x) \cdot dx + \cdots + \int_{b-4h}^{b-2h} f(x) \cdot dx + \int_{b-2h}^b f(x) \cdot dx \\ &= \int_{x_0}^{x_2} f(x) \cdot dx + \int_{x_2}^{x_4} f(x) \cdot dx + \cdots + \int_{x_{N-4}}^{x_{N-2}} f(x) \cdot dx + \int_{x_{N-2}}^{x_N} f(x) \cdot dx \end{aligned}$$

$$\begin{aligned} a &= x_0 \\ b &= x_N \end{aligned} \rightarrow h = \frac{b-a}{N} = x_1 - x_0 = x_2 - x_1 = \cdots = x_N - x_{N-1}$$

$$\begin{aligned} R_{[a,b]}^{(2)} &= R_{[x_0,x_2]}^{(2)} + R_{[x_2,x_4]}^{(2)} + \cdots + R_{[x_{N-2},x_N]}^{(2)} = -\frac{h^5}{90} \cdot f^{(4)}(\xi_1) - \frac{h^5}{90} \cdot f^{(4)}(\xi_2) - \cdots - \frac{h^5}{90} \cdot f^{(4)}(\xi_{N/2}); \\ &= -\frac{(b-a) \cdot h^4}{180} \cdot f^{(4)}(\xi); \quad x_0 \leq \xi \leq x_N \end{aligned}$$



$$\begin{aligned} x_0 &\leq \xi_1 \leq x_2 \\ x_2 &\leq \xi_2 \leq x_4 \\ &\vdots \\ x_{N-2} &\leq \xi_{N/2} \leq x_N \end{aligned}$$